Objectness Gate Change for the HiCE Object-of-Interest Branch

Short implementation note: separating **where to look** from **whether to trust the object feature**.

1. Current plan

The current object branch predicts a 7x7 heatmap over the post-hand-cross-attention feature map. The heatmap is converted to a spatial attention distribution, used to pool an object feature, and then the pooled feature is gated by the maximum heatmap logit.

Current equations:

A = softmax(H.flatten())
f_obj = sum_i A_i F_i
gate = sigmoid(max(H))
f_obj_gated = gate * f_obj

The motivation is correct: if no candidate object is visible in the current crop, softmax still produces a distribution over the 49 cells and therefore pools some background feature. The gate is intended to shrink the object feature when the heatmap has low confidence.

2. Why sigmoid(max(H)) can be brittle

The issue is that max(H) is being used for two different meanings at the same time: localization confidence and object presence. A heatmap should answer *where should I pool?*. Objectness should answer *should I trust the pooled object feature at all?*. These are related but not the same.

Failure mode	**What happens with sigmoid(max(H))**	**Why it is bad**
Wrong sharp peak	One background cell gets a high logit, so max(H) is high and gate ~= 1.	The model injects a strong but wrong object feature into HiCE.
Correct but conservative heatmap	The heatmap location is reasonable, but logits are small early in training, so gate is low.	The object feature is suppressed, and event loss cannot benefit much from it.
No object in crop	Softmax still pools some cells; max(H) may become high due to noise.	The branch may add background garbage instead of reverting to vanilla HiCE.

3. Proposed change: separate object presence gate

Keep the heatmap for spatial attention, but add a tiny scalar presence head q_obj. This head predicts whether a useful candidate object exists in the current crop. The heatmap says where to read; the presence gate says how much to trust the readout.

New equations:

A = softmax(H.flatten())
f_obj = sum_i A_i F_i
q_obj = sigmoid(MLP([f_global, f_hand]))
f_obj_gated = q_obj * f_obj

This is cleaner because the network has an explicit supervised signal for the all-zero target case. If the rendered object target has no positive cells, q_obj should be 0. If a candidate object is visible/relevant in the crop, q_obj should be 1.

4. Supervision

The heatmap loss remains the same. Add one scalar BCE for object presence.

Targets:

y_obj = 1 if the rendered object target has any positive cell
y_obj = 0 if the target is all zero, meaning no candidate object is visible in the crop

Loss:

L_obj = L_heatmap + lambda_presence * BCE(q_obj, y_obj)
L_total = L_HiCE + lambda_obj * L_obj

Suggested starting values: lambda_obj = 0.1 to 0.2, lambda_presence = 0.5 to 1.0 inside L_obj. Keep the object branch light so it does not dominate the event detection loss.

5. Recommended implementation

The smallest implementation is a presence MLP that reads the global feature and, if available, the hand feature. It does not require a detector, NMS, box decoding, or a new temporal loss.

Pseudo-code:

```
# F: (B*L, D, 7, 7) hand-aware feature map
# f_global: (B*L, D) pooled frame feature
# f_hand: (B*L, D) hand feature or left/right summary
H = obj_heatmap_head(F) # (B*L, 1, 7, 7)
A = softmax(H.flatten(2), dim=-1) # spatial attention
f_obj = (F.flatten(2) * A).sum(-1) # (B*L, D)
q_obj = sigmoid(presence_mlp(cat(f_global, f_hand)))
f_obj = q_obj * f_obj
f_out = f_global + gamma * proj(f_obj) # gamma initialized to 0
```

Using residual fusion with gamma initialized to zero is safer than replacing the feature with Linear(concat(f_global, f_obj)). At step 0, the model is exactly vanilla HiCE; the object branch only contributes once it learns something useful.

6. Why this is the better design

Current gate: sigmoid(max(H))	Separate presence gate: q_obj
Presence is inferred indirectly from one maximum heatmap logit.	Presence is explicitly predicted and supervised.
A wrong sharp peak can produce high confidence.	A wrong location can still be downweighted if object presence is low.

Hard to interpret whether failure is heatmap location or object confidence.	Debuggable: report heatmap quality and q_obj accuracy separately.
Can inject background garbage when the crop misses the object.	Can learn to suppress object feature in all-zero target frames.

7. Practical recommendation

For a quick prototype, sigmoid(max(H)) is acceptable because it is very simple. For the final method, use the separate presence gate. It gives a cleaner story: the heatmap learns where the candidate object is, while the presence gate learns whether that object feature should be trusted. This is especially important because the hand-based crop can exclude the object in some frames.

Suggested ablation:

1) HiCE baseline
2) + heatmap loss only
3) + heatmap attention with sigmoid(max(H)) gate
4) + heatmap attention with separate q_obj presence gate
5) + residual fusion initialized at zero